FLAGSHIP 08 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-ACC/W

Compressor Water Cooling Vest · 300W Extreme Micro-Inverter Chiller

300W Active Micro-Inverter Refrigeration for Extreme Heat Hazards.

The flagship of the LC Frost line, the LC Frost-ACC/W integrates a miniature inverter refrigeration compressor with a closed-loop chilled liquid circulation network. Delivering an industry-leading 60-300W of active cooling capacity, a wide temperature adjustment range, and unwavering constant-temperature output, it is the ultimate protective cooling armor for extreme metallurgy, firefighting, and confined-space operations.

01 300W Extreme Active Refrigeration

Micro-inverter compressor rapidly chills circulating water upon startup with multi-stage digital temperature regulation.

02 Under-4kg Integrated Mobility

Compact cooling architecture balances weight along the lower torso for unrestricted mobility in tight quarters.

03 Intensive 2-4 Hour Shift Endurance

High-capacity 24V lithium battery system sustains heavy-duty refrigeration across grueling shifts.

04 Shockproof & Sealed Fluidics

Rugged liquid couplings and pressurized TPU bladders withstand heavy impacts, vibration, and site washdowns.

LC Frost-ACC/W · 3200K STUDIO MASTER



KEY METRIC 01
60-300W
Heavy-Duty Active Compressor Refrigeration

KEY METRIC 02
< 4kg
Ergonomically Balanced Wearable Chiller Unit

KEY METRIC 03
3 min
Rapid High-Capacity Chilled Loop Stabilization

KEY METRIC 04
100%
Sealed Leak-Proof Liquid Circuit

MATERIAL SCIENCE & MICRO-ARCHITECTURE · 3200K STUDIO MACRO INSPECTION



60·300W MICRO-INVERTER COMPRESSOR

24V Miniature Inverter Refrigeration Compressor Core

Unlike standard thermoelectric coolers, the LC Frost-ACC/W utilizes a true vapor-compression micro-inverter cycle capable of delivering up to 300W of refrigeration power - maintaining cold liquid circulation even in 50°C+ foundry radiation.



PRESSURIZED TPU LIQUID BLADDER

High-Durability Pressurized TPU Liquid Cooling Bladder

High-density serpentine TPU cooling bladder maximizes torso contact surface area while an inner breathable 3D mesh barrier prevents skin clamminess.



ANTISTATIC FR SHELL · < 4KG TOTAL

Antistatic Flame-Retardant Shell & Heavy-Duty Center Zipper

Industrial grey antistatic and flame-retardant outer fabric with reinforced black trim and bilateral compression straps resists sparks, slag, and heavy abrasion.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **Flame-Retardant V-Neck Outer Shell [Antistatic FR V-Neck Armor Shell]**
   Minimalist grey antistatic shell with heavy-duty center zipper fits comfortably over base layers.

2. **Bilateral Compression Webbing [Bilateral Torso Compression Straps]**
   Side tension straps maximize thermal contact between the internal chilled TPU bladder and torso.

3. **Micro-Inverter Compressor Chiller [60-300W Micro-Inverter Refrigeration Core]**
   24V miniature compressor core stabilizes chilled water circulation within 3 minutes.

4. **Internal TPU Cooling Bladder & Insulated Loop [Pressurized Closed-Loop TPU Bladder]**
   Closed-loop liquid network paired with breathable 3D lining prevents condensation accumulation.

FIELD DEPLOYMENT | BLAST FURNACE & MOLTEN METAL CASTING BAY | >45°C AMBIENT



Extreme Metallurgy Foundries, Firefighting & Confined-Space Operations

In blast furnace casting bays, heavy equipment cabs without HVAC, and deep confined-space rescue missions, radiant heat loads overwhelm conventional cooling vests. Delivering 60-300W of active vapor-compression refrigeration in a <4kg wearable package, the LC Frost-ACC/W stabilizes chilled water within 3 minutes and protects core body temperature under the most punishing industrial conditions on earth.

60-300W (24V / 110W)	3 Minutes	< 4.0kg
Active Refrigeration Range	Chilled Loop Stabilization	Complete System Weight

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-ACC/W (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-ACC/W
IDENTITY & COMPLIANCE	**Color**	Industrial Grey with Black Trim
IDENTITY & COMPLIANCE	**Target Users**	Extreme high-temperature environments (metallurgy, foundry, firefighting, heavy equipment cabs)
PERFORMANCE & MECHANICS	**Cooling Capacity**	60-300W
PERFORMANCE & MECHANICS	**Rated Voltage**	24V
PERFORMANCE & MECHANICS	**Rated Power**	110W
PERFORMANCE & MECHANICS	**Cooling Response Time**	Stable in 3 min
PERFORMANCE & MECHANICS	**Battery Life**	1-2 hours single battery (2-4 hours system continuous operation)
MATERIALS & ERGONOMICS	**Outer Fabric**	Antistatic and flame-retardant fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	High-durability TPU liquid cooling bladder
DIMENSIONS & WARRANTY	**Size**	One Size Fits All
DIMENSIONS & WARRANTY	**Net Weight**	< 4kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-ACC/W

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware